

LAB 3: Simulation of Network Devices using Cisco Packet Tracer

OBJECTIVE

- To understand the basic concepts of computer networking and network devices.
- To simulate and configure common network devices such as switches, hubs, bridge, routers, repeaters and end devices.
- To observe data transmission between devices in a simulated network environment.

THEORY

Computer networks are formed by interconnecting multiple computing devices so that they can communicate with one another and share data, software, and hardware resources such as printers and internet connections. These interconnected devices exchange information using defined communication rules called protocols. Network devices such as hubs, switches, routers, bridges, and repeaters are essential components of a network because they manage how data is transmitted, forwarded, and controlled. Each device performs a specific function to ensure that data reaches the correct destination efficiently and reliably.

Cisco Packet Tracer

Cisco Packet Tracer is a popular graphical network simulation software developed by Cisco Systems. It enables students and professionals to design, configure, and test computer networks in a virtual environment without requiring physical hardware. Packet Tracer supports a wide range of networking devices including routers, switches, hubs, wireless access points, and end devices such as personal computers and servers. The software provides two working modes: Real-Time Mode, which shows network operation as it would occur in real life, and Simulation Mode, which allows users to observe packet flow step by step. This visualization helps users understand how protocols work and how data travels through a network, making Packet Tracer an effective learning tool.

Hub

A hub is a basic networking device used to connect multiple computers in a local area network. It works at the Physical Layer (Layer 1) of the OSI model. When a hub receives data from one device, it does not check the destination address. Instead, it broadcasts the data to all connected devices, which leads to unnecessary traffic. Because all devices share the same bandwidth, data collisions occur frequently, reducing network efficiency. Hubs do not store MAC addresses and do not provide any security features. They support only half-duplex communication. Due to poor performance and high collision rates, hubs are rarely used in modern networks and have mostly been replaced by switches.



Switch

A switch is an intelligent networking device used to connect multiple devices within a local area network. It operates at the Data Link Layer (Layer 2) of the OSI model. A switch forwards data by examining the MAC address of the destination device and sends the data only to the correct port. This reduces unnecessary traffic and minimizes collisions. Switches support full-duplex communication, allowing devices to send and receive data simultaneously. They improve network performance, efficiency, and security compared to hubs. Switches maintain a MAC address table to track connected devices. Due to these advantages, switches are widely used in homes, offices, and enterprise networks.



Bridge

A bridge is a networking device used to divide a large network into smaller segments to improve performance. It works at the Data Link Layer (Layer 2) of the OSI model. A bridge filters network traffic by examining MAC addresses and decides whether to forward or block data frames. By reducing unnecessary traffic between segments, bridges help minimize collisions and improve overall network efficiency. Bridges learn MAC addresses dynamically and store them in a table. However, bridges are slower than switches because they have fewer ports and limited processing power. In modern networks, bridges are mostly replaced by switches, which perform similar functions more efficiently.



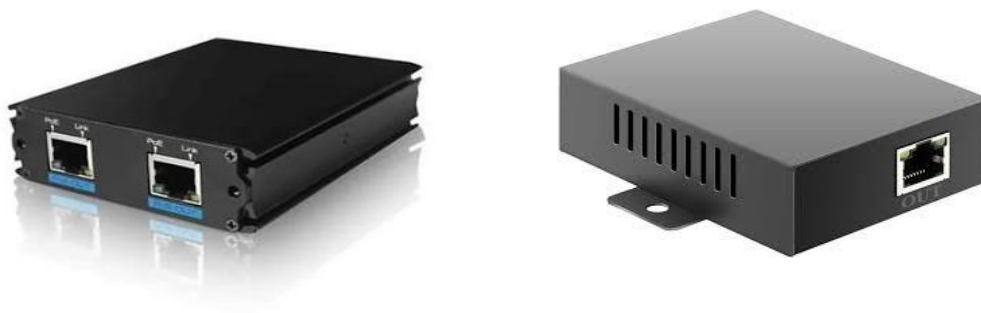
Router

A router is a networking device used to connect different networks and forward data between them. It operates at the Network Layer (Layer 3) of the OSI model. Routers use IP addresses to determine the best path for data transmission. They maintain routing tables to select the most efficient route for packets. Routers are commonly used to connect local area networks to wide area networks and the Internet. They also provide advanced features such as traffic management, network security, and access control. By separating broadcast domains, routers reduce network congestion and enhance performance. Routers are essential components of modern networking infrastructures.



Repeater

A repeater is a networking device used to extend the distance of a network. It operates at the Physical Layer (Layer 1) of the OSI model. When data signals travel over long distances, they become weak or distorted. A repeater receives these weak signals, regenerates them, and retransmits them with original strength. Repeaters do not understand data frames or addresses and do not filter traffic. They simply amplify and reshape signals. Repeaters are useful in long cable networks to overcome signal attenuation. However, they do not reduce network traffic or collisions and are mainly used for signal strengthening purposes.



In this lab, Cisco Packet Tracer was used to simulate and study the operation of basic network devices. Devices such as hub, switch, bridge, router, and repeater were placed in a virtual network and interconnected using appropriate cables. The configuration and functioning of each device were observed to understand their roles in data transmission and network communication. By sending data packets between end devices, the behavior of these network components in managing traffic, forwarding data, and maintaining connectivity was analyzed. This practical simulation helps in gaining hands-on experience and reinforces theoretical networking concepts. The following sections explain the working principles, functions, and characteristics of each network device used in the experiment.

Network Topology

1. HUB

HUB: Hub-PT Hub0

PC Name	IP Address
PC-PT PC0	192.168.1.1
PC-PT PC1	192.168.1.2
PC-PT PC2	192.168.1.3
PC-PT PC3	192.168.1.4
PC-PT PC4	192.168.1.5
PC-PT PC5	192.168.1.6

PC0

Physical Config Desktop Programming Attributes

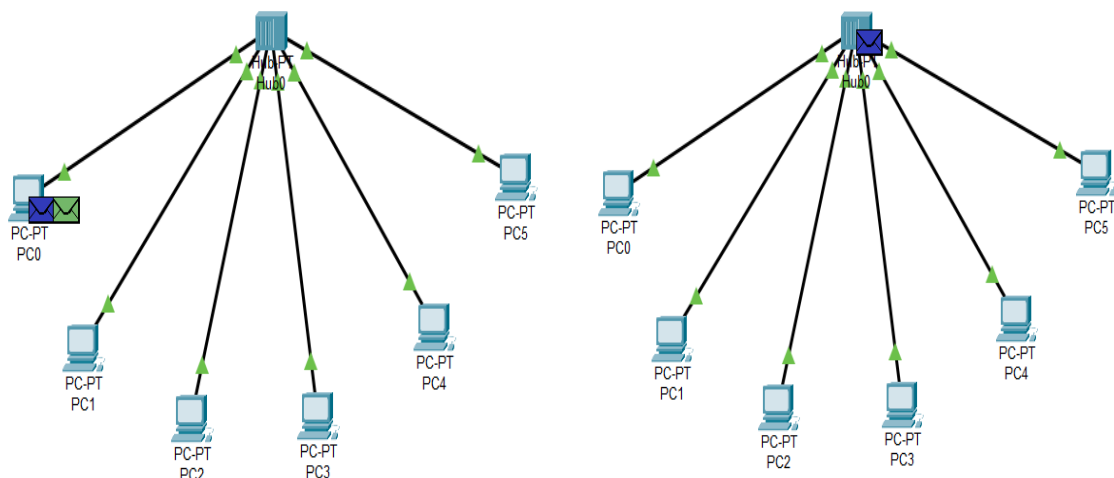
Command Prompt

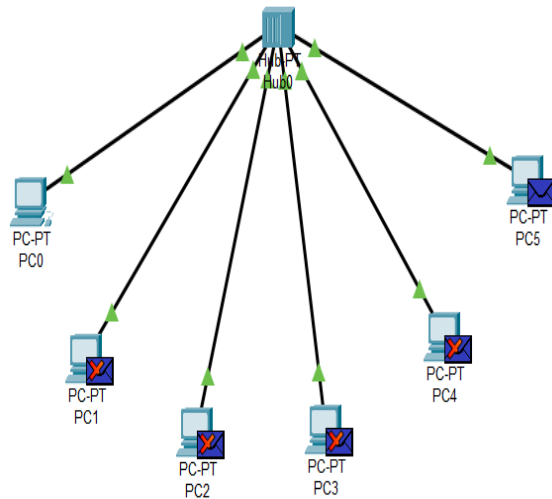
```
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.6

Pinging 192.168.1.6 with 32 bytes of data:

Reply from 192.168.1.6: bytes=32 time=8ms TTL=128
Reply from 192.168.1.6: bytes=32 time=4ms TTL=128
Reply from 192.168.1.6: bytes=32 time=4ms TTL=128
Reply from 192.168.1.6: bytes=32 time=4ms TTL=128

Ping statistics for 192.168.1.6:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 8ms, Average = 5ms
```





Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC0	ICMP
	0.001	PC0	Hub0	ICMP
	0.002	Hub0	PC1	ICMP
	0.002	Hub0	PC2	ICMP
	0.002	Hub0	PC3	ICMP
	0.002	Hub0	PC4	ICMP
	0.002	Hub0	PC5	ICMP

Discussion: When data was sent from one PC connected to a hub, the packet was broadcast to all connected devices. In the simulation, every PC received the frame, but only the intended destination accepted it, while others discarded it. Multiple PCs trying to send data caused collisions, which were visible in Packet Tracer. This showed that the hub does not filter traffic and works inefficiently.

2. Switch

SWITCH: 2960-24TT Switch3

PC Name	IP Address
PC-PT PC13	192.168.1.1
PC-PT PC14	192.168.1.2
PC-PT PC15	192.168.1.3
PC-PT PC16	192.168.1.4
PC-PT PC17	192.168.1.5
PC-PT PC18	192.168.1.6

```

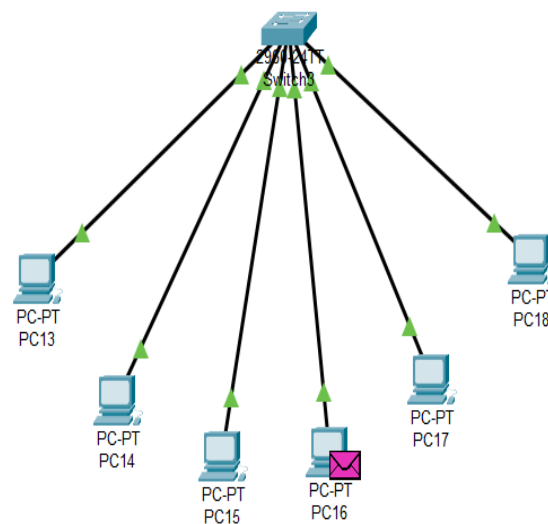
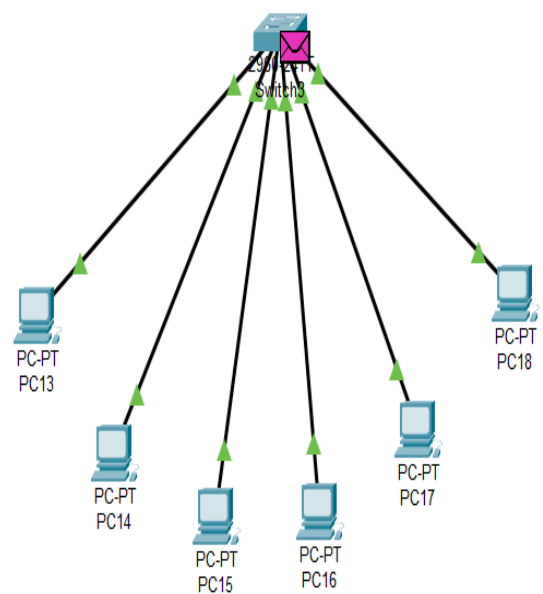
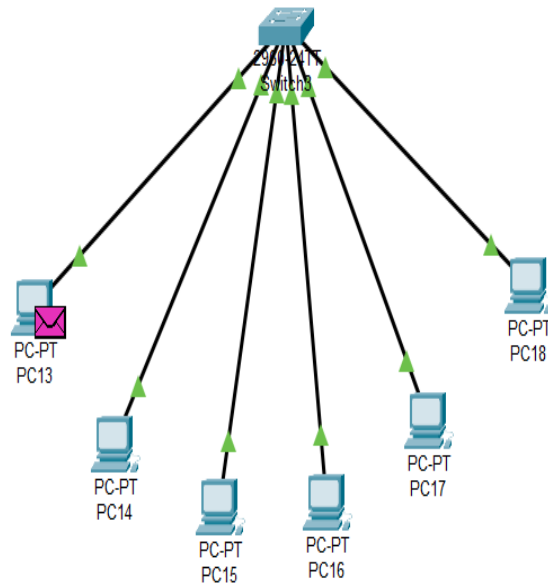
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:

Reply from 192.168.1.4: bytes=32 time=8ms TTL=128
Reply from 192.168.1.4: bytes=32 time=4ms TTL=128
Reply from 192.168.1.4: bytes=32 time=4ms TTL=128
Reply from 192.168.1.4: bytes=32 time=4ms TTL=128

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 8ms, Average = 5ms

```



Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC13	ICMP
	0.001	PC13	Switch3	ICMP
	0.002	Switch3	PC16	ICMP

Discussion: In the switch-based topology, data sent from a source PC was forwarded only to the destination PC. Initially, the switch flooded the frame to learn MAC addresses, but after that, packets followed a single correct path. No unnecessary traffic or collisions were observed. This output confirmed that the switch uses a MAC address table and provides efficient communication.

3. Bridge

BRIDGE: Bridge-PT Bridge0

SWITCH: 2960-24TT Switch4

2960-24TT Switch5

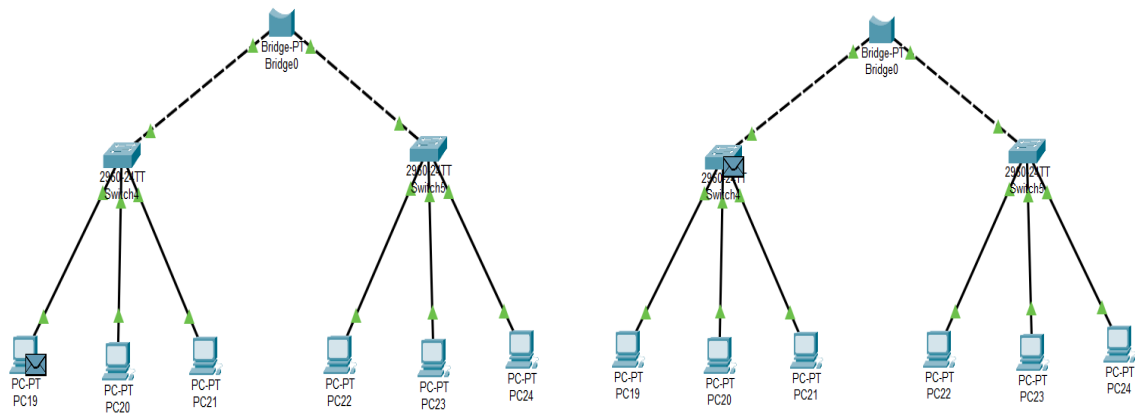
PC Name	IP Address
PC-PT PC19	192.168.1.1
PC-PT PC20	192.168.1.2
PC-PT PC21	192.168.1.3
PC-PT PC22	192.168.1.4
PC-PT PC23	192.168.1.5
PC-PT PC24	192.168.1.6

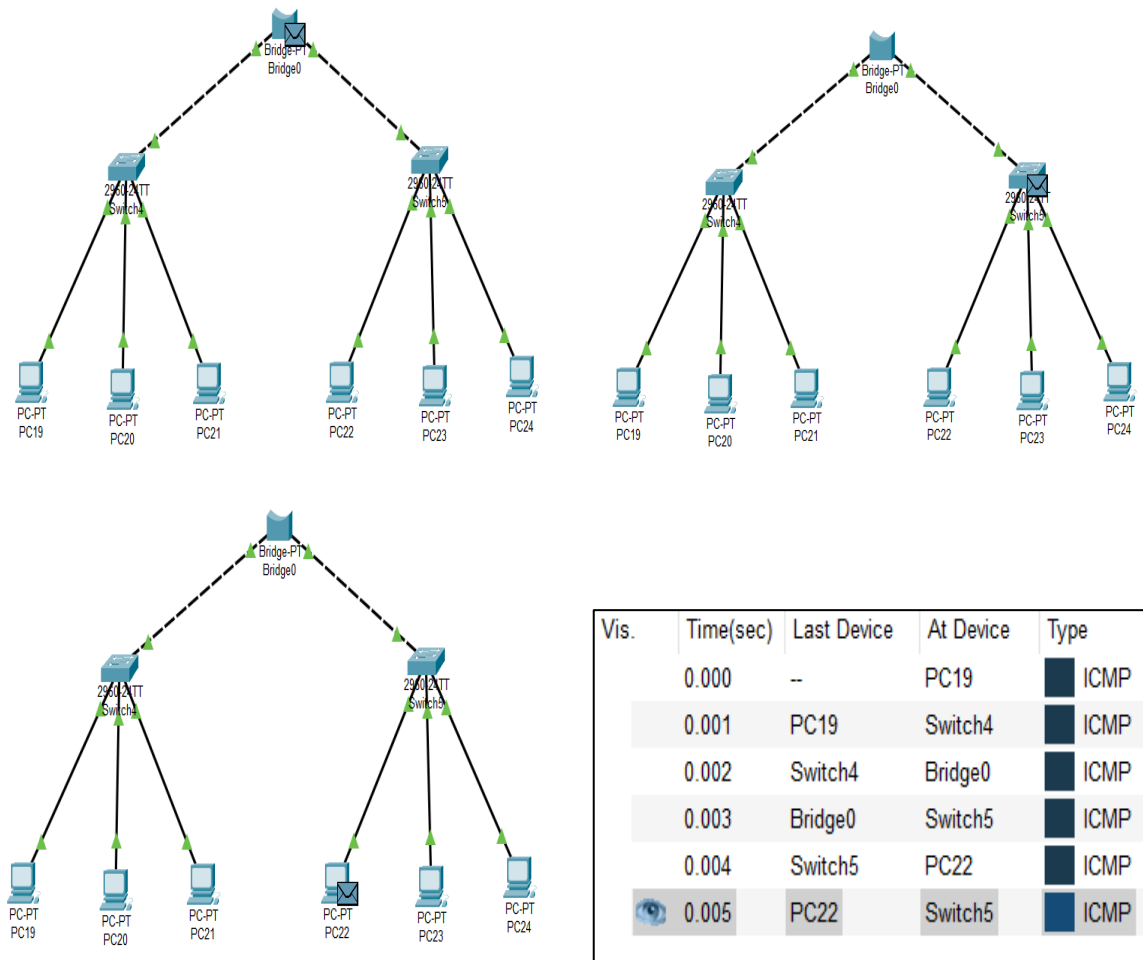
```
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:

Reply from 192.168.1.4: bytes=32 time=16ms TTL=128
Reply from 192.168.1.4: bytes=32 time=8ms TTL=128
Reply from 192.168.1.4: bytes=32 time=8ms TTL=128
Reply from 192.168.1.4: bytes=32 time=8ms TTL=128

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 8ms, Maximum = 16ms, Average = 10ms
```





Discussion: When a bridge was used, data frames were filtered based on MAC addresses. Packets were forwarded only to the network segment containing the destination device. Devices on other segments did not receive the traffic. The simulation showed reduced collisions compared to a hub, proving that the bridge successfully separates network segments.

4. Router

SWITCH: 2960-24TT Switch4

2960-24TT Switch5

Router: 1941 Router2	IP Address
GigabitEthernet0/0	192.168.1.1
GigabitEthernet0/1	10.10.10.1

PC Name	IP Address
PC-PT PC25	192.168.1.2
PC-PT PC26	192.168.1.3
PC-PT PC27	192.168.1.4
PC-PT PC28	10.10.10.2
PC-PT PC29	10.10.10.3
PC-PT PC30	10.10.10.4

```

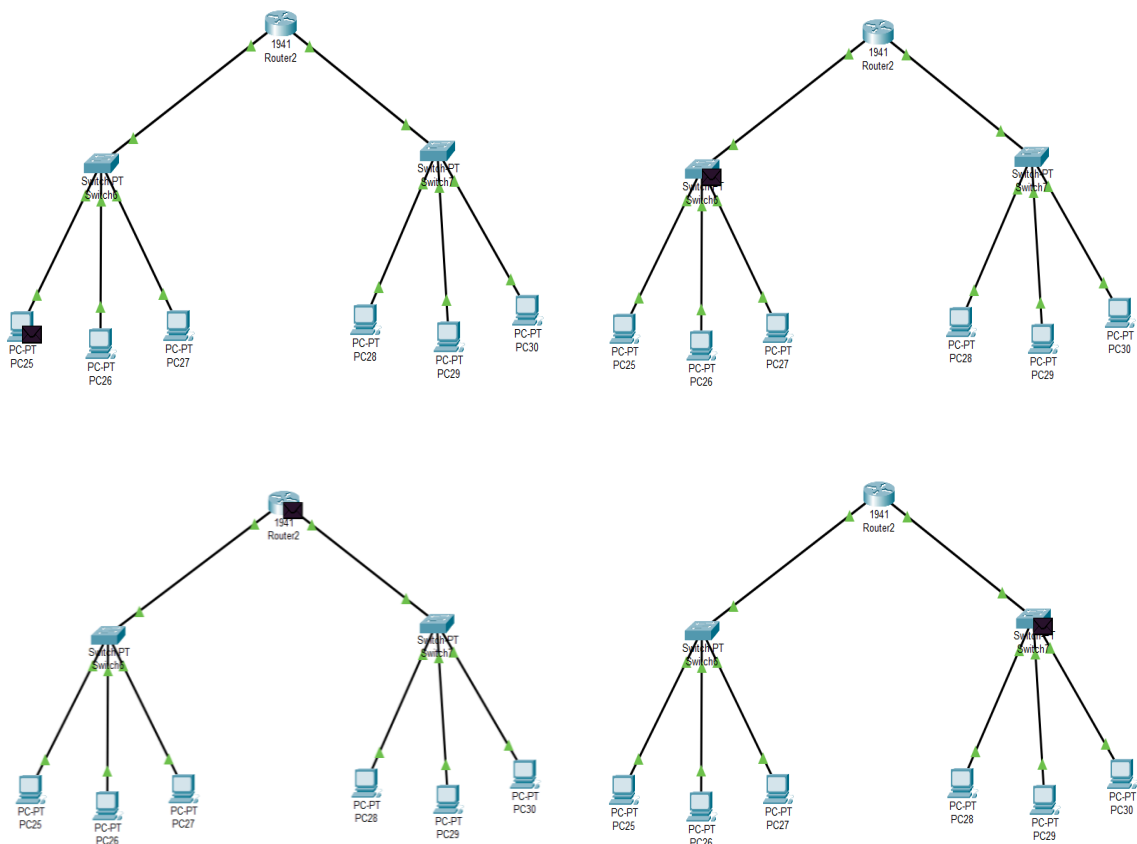
C:\>ping 10.10.10.2

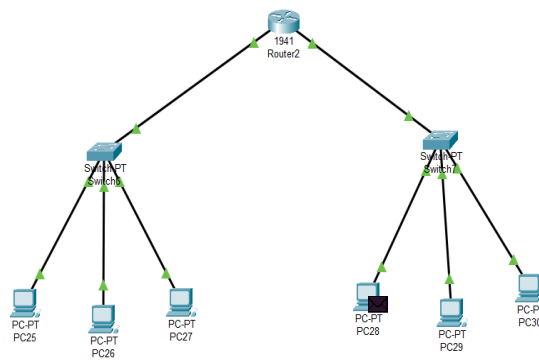
Pinging 10.10.10.2 with 32 bytes of data:

Request timed out.
Reply from 10.10.10.2: bytes=32 time=8ms TTL=127
Reply from 10.10.10.2: bytes=32 time=8ms TTL=127
Reply from 10.10.10.2: bytes=32 time=8ms TTL=127

Ping statistics for 10.10.10.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 8ms, Maximum = 8ms, Average = 8ms

```





Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC25	ICMP
	0.001	PC25	Switch6	ICMP
	0.002	Switch6	Router2	ICMP
	0.003	Router2	Switch7	ICMP
	0.004	Switch7	PC28	ICMP

Discussion: In the router topology, packets were forwarded using IP addresses instead of MAC addresses. The router allowed communication only between different networks when proper IP configuration was done. Broadcast packets were not forwarded, which reduced network congestion. The output clearly showed routing decisions and packet movement between separate networks.

5. Repeater

REPEATER: Repeater-PT Repeater0

SWITCH: 2960-24TT Switch3

2960-24TT Switch4

PC Name	IP Address
PC-PT PC0	192.168.1.1
PC-PT PC1	192.168.1.2
PC-PT PC2	192.168.1.3
PC-PT PC3	192.168.1.4
PC-PT PC4	192.168.1.5
PC-PT PC5	192.168.1.6

```

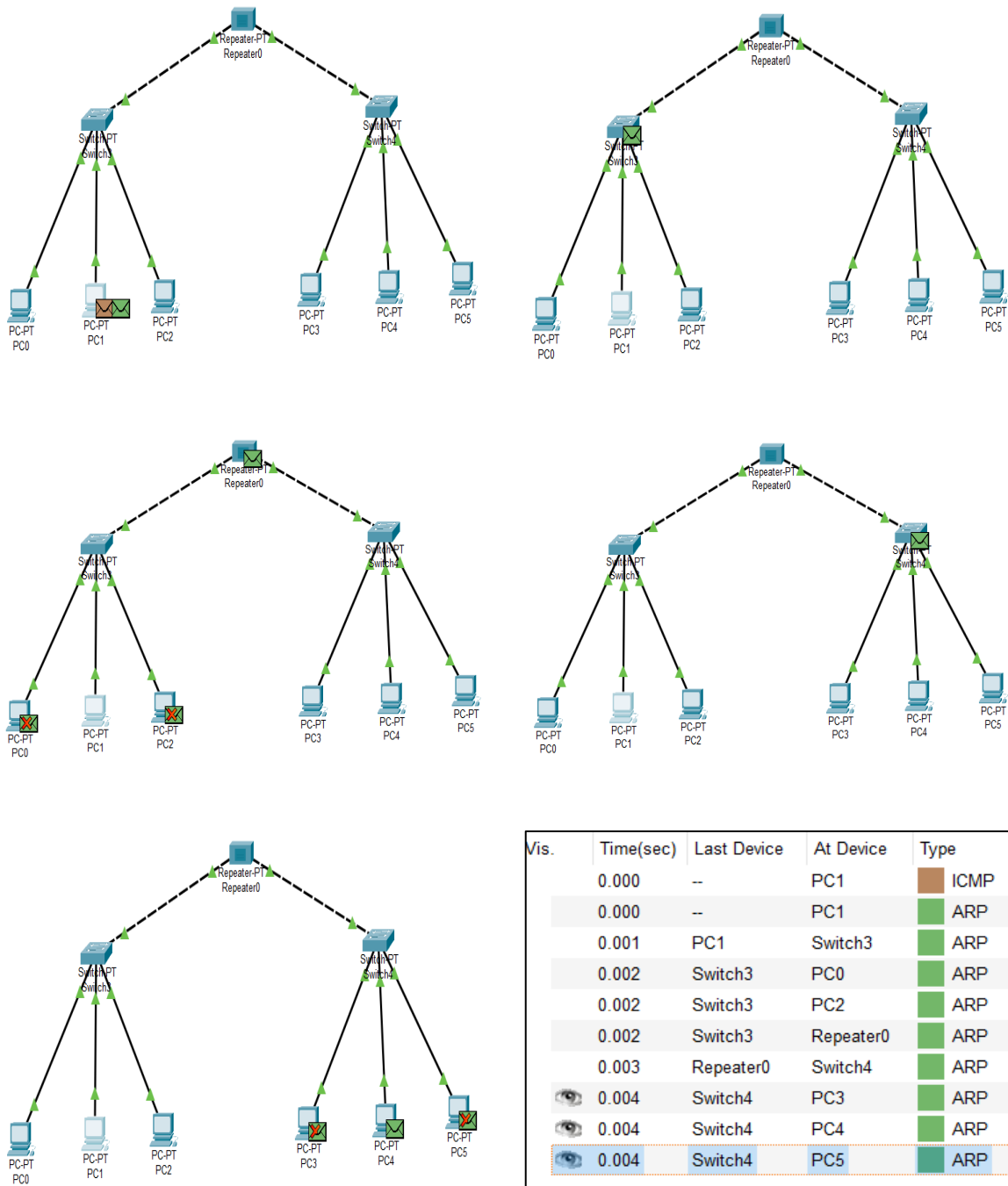
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.5

Pinging 192.168.1.5 with 32 bytes of data:

Reply from 192.168.1.5: bytes=32 time=16ms TTL=128
Reply from 192.168.1.5: bytes=32 time=8ms TTL=128
Reply from 192.168.1.5: bytes=32 time=8ms TTL=128
Reply from 192.168.1.5: bytes=32 time=8ms TTL=128

Ping statistics for 192.168.1.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 8ms, Maximum = 16ms, Average = 10ms

```



Discussion: When a repeater was placed between two network segments, the signal was regenerated and retransmitted. The packet flow remained the same, but the signal successfully reached longer distances without loss. The repeater did not filter or modify data, only strengthened it. This confirmed that repeaters operate only at the physical layer.

DISCUSSION AND CONCLUSION

This experiment successfully demonstrated the working behavior of basic network devices using Cisco Packet Tracer. By observing packet flow in different topologies, the functional differences between hub, switch, bridge, router, and repeater were clearly understood. The hub broadcasted data to all devices, causing collisions, while the switch and bridge reduced unnecessary traffic by using MAC addresses. The router enabled communication between different networks using IP addressing and controlled broadcasts. The repeater improved signal strength without filtering data.

Overall, the simulation reinforced theoretical concepts and provided practical understanding of data transmission and device roles in computer networks.